

MOUNT WILSON, CA, USA

WMO: 722890

Lat: 34.232N

Lon: 118.074W

Elev: 5706

StdP: 11.91

Time Zone: -8.00 (NAP)

Period: 82-15

WBAN: 93136

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	24.5	26.9	-18.1	2.5	51.9	-12.9	3.4	51.6	24.7	36.8	20.9	38.4	6.5	320	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	88.2	59.6	85.9	59.2	82.5	57.8	64.1	80.8	62.5	80.0	61.1	78.8	5.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	90.2	71.0	55.5	81.1	70.8	53.9	76.5	71.3	32.4	80.6	31.1	80.4	30.0	78.5	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	14.9	12.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	47.0	46.9	47.6	52.1	59.0	65.1	72.6	72.0	69.6	59.3	50.6	44.2		
	DBStd	13.24	8.76	9.67	9.26	9.76	9.58	8.34	5.35	5.92	7.26	9.29	9.16	9.24		
	HDD50	983	160	160	159	92	37	8	0	0	3	30	106	228		
	HDD65	3682	557	508	540	396	228	94	8	13	41	218	435	644		
	CDD50	3621	68	73	84	155	315	461	700	682	589	320	124	50		
	CDD65	844	0	0	0	8	41	98	243	229	178	43	4	0		
	CDH74	5960	0	2	2	40	321	693	1785	1750	1132	221	14	0		
	CDH80	1483	0	0	0	2	35	165	466	490	297	28	0	0		
(14) Wind	WSAvg	5.2	5.5	6.0	5.9	5.9	5.0	4.7	4.2	4.2	4.8	4.7	5.5	5.5		
(15) Precipitation	PrecAvg	19.8	4.3	4.8	2.5	1.4	0.6	0.2	0.0	0.0	0.2	0.8	1.6	3.0		
	PrecMax	37.4	18.1	16.8	10.9	3.3	5.3	1.2	0.5	0.1	2.2	6.6	4.3	11.2		
	PrecMin	8.1	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.3		
	PrecStd	8.6	4.7	4.3	2.8	0.9	1.0	0.3	0.1	0.0	0.5	1.3	1.1	2.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	72.0	72.4	78.7	83.7	89.8	90.2	91.0	90.1	82.3	76.1	70.5		
		MCWB	45.9	47.6	49.9	52.0	55.1	58.9	61.1	62.2	58.7	53.9	49.7	47.7		
	2%	DB	65.1	66.7	68.2	72.9	80.9	83.9	87.6	88.1	86.2	79.1	71.1	64.3		
		MCWB	43.3	45.4	47.9	49.8	55.4	56.6	60.4	59.9	58.3	53.4	47.6	44.0		
	5%	DB	62.4	63.2	63.9	69.9	77.7	80.7	84.4	85.1	82.8	75.7	66.5	61.1		
		MCWB	42.1	44.5	46.6	48.8	53.8	55.9	59.5	59.5	57.4	51.8	46.5	42.1		
	10%	DB	58.5	59.8	60.5	66.5	73.9	77.5	81.9	82.0	80.8	72.4	63.0	56.8		
		MCWB	39.8	42.2	44.1	47.4	52.5	54.5	58.4	58.2	56.3	49.8	43.5	40.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	50.4	52.9	55.4	60.4	62.9	66.1	65.4	64.8	59.1	51.5	50.8		
		MCDB	61.9	63.0	65.8	70.6	77.5	82.3	81.9	80.5	79.1	74.8	68.2	62.4		
	2%	WB	45.7	47.8	50.3	52.8	58.0	60.3	64.3	63.8	62.3	56.1	49.9	47.8		
		MCDB	59.1	61.8	64.1	67.1	75.0	79.4	80.9	81.4	79.0	72.5	66.5	58.7		
	5%	WB	43.7	45.8	47.9	50.6	56.0	58.4	62.6	62.2	60.2	54.0	47.9	45.1		
		MCDB	56.1	60.2	62.0	66.1	74.0	76.6	79.8	80.3	77.6	70.1	63.5	53.6		
	10%	WB	41.9	43.9	45.6	48.7	54.3	56.3	61.1	60.8	58.7	51.8	45.9	42.6		
		MCDB	54.1	57.0	58.5	64.3	72.0	74.9	78.4	78.7	76.4	68.5	58.4	53.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	14.7	15.1	15.9	19.6	15.5	14.9	16.3	16.2	16.2	14.4	14.2		
		MCDBR	18.4	18.9	21.0	18.8	22.0	18.6	17.6	18.7	20.4	19.3	19.0	17.9		
		MCWBR	11.6	13.2	15.3	13.4	15.9	14.1	13.9	12.7	12.9	12.5	12.0	12.6		
	5% WB	MCDBR	16.0	17.5	19.3	19.0	21.2	18.8	15.6	16.3	17.9	18.1	17.9	16.1		
		MCWBR	11.5	13.9	15.8	15.6	17.4	16.6	13.4	13.0	13.1	14.3	13.1	13.3		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.267	0.282	0.287	0.288	0.301	0.303	0.292	0.282	0.284	0.253	0.252		
	taud		2.618	2.619	2.551	2.539	2.569	2.531	2.579	2.602	2.626	2.588	2.683	2.672		
	Ebn at Noon		309	317	317	317	315	310	308	312	313	305	310	307		
	Edh at Noon		23	26	30	32	32	33	31	30	28	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		1004	1288	1737	2127	2346	2513	2437	2304	1954	1483	1128	914		
	RadStd		74	133	95	102	130	130	114	65	59	93	56	66		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	+1.15	N/A	-2.99	+1.28	N/A	N/A	N/A	N/A	-205	+405	+198

Nomenclature: See separate page